

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=hard_cheese

retailer=silpo_novus

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	0.0016491839418000672	0.01	1.66813437649029e-12	0.1	4.160589056408487e-21	0.1	I(1)	1.0	hard_cheese	silpo_novus	promo_controlled	nan	nan
prozorro	2.633366148124232e-10	0.08893253878307611	1.5324738060670524e-08	0.1	1.1965567303232429e-10	0.041666666666668085	I(0)	0.0	hard_cheese	silpo_novus	promo_controlled	nan	nan
retail	0.00963902009133378	0.1	1.9158401072775886e-24	0.1	0.0012553169947406746	0.1	I(0)	0.0	hard_cheese	silpo_novus	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	hard_cheese	silpo_novus	promo_controlled	0.23227008326013393	189.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	hard_cheese	silpo_novus	promo_controlled	0.00039637380760688387	49.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	hard_cheese	silpo_novus	promo_controlled	0.9658005654865729	57.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
hard_cheese	silpo_novus	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	187	-1.901684280673745	0.31953965595429906	-0.703887241265392	2.747609431022365e-15	0.16253431119720008	0.2610022514748195	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
hard_cheese	silpo_novus	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	49	-1.3314737468808624	-3.3232402194259745	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=0.5652; bo... Primary chain is strictly...	
hard_cheese	silpo_novus	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	47	-0.286917649947797	-7.00476787473343	-0.5857286396943655	2.5559001487149504e-09	0.8452940711711397	0.015014208714442796	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
NARDL	producer_to_prozorro	prozorro	producer	0.30453926702142564	0.2688968463348229	0.05424410826903407	0
ARDL	prozorro_to_retail	retail	prozorro	0.8676166313512811	0.46209891936694814	0.2508240957204422	0
NARDL	prozorro_to_retail	retail	prozorro	0.9705702295552536	0.49641237836935825	0.15198175860587926	0

Table: SeriesUsed

	date	producer	prozorro	retail	standardized	type	retailer	promo_variant	combined_rule
	2021-02-01	32.03526456031363	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-02	32.04618683176164	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-03	32.05676642836296	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-04	32.06697823935806	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-05	32.07679715398746	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-06	32.08619806149162	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-07	32.09515585111103	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-08	32.10364541208618	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-09	32.11123400633844	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-10	32.1178387696982	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-11	32.12395940226944	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-12	32.13009560415616	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-13	32.13674707546236	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-14	32.14441351629204	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-15	32.15359462674919	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-16	32.16772969300264	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-17	32.18792409272353	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-18	32.21117666326934	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-19	32.23448624199752	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-20	32.25485166626554	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-21	32.26927177343089	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-22	32.274745400851	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-23	31.92798797934096	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-24	31.03371883965718	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-25	29.81094266906914	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-26	28.47866415484636	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-27	27.25588798425832	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-02-28	26.36161884457453	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-01	26.01486142306449	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-02	26.03157873440422	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-03	26.07693453328602	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-04	26.14373461700382	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-05	26.22478478285154	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-06	26.31289082812309	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-07	26.40085855011242	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-08	26.48149374611343	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-09	26.56908821932747	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-10	26.67500355097146	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-11	26.78765020536267	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	
	2021-03-12	26.89543864681836	nan	nan	hard_cheese	silpo_novus	promo_controlled	daily_median_of_available...	

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1	0.05463411003148463	
producer_to_prozorro	2	0.06465611259680928	
producer_to_prozorro	3	0.0701974599652885	
producer_to_prozorro	4	0.07719408938724882	
producer_to_prozorro	5	0.07629421025159185	
producer_to_prozorro	6	0.07219617159882989	
producer_to_prozorro	7	0.07415560821992007	
producer_to_prozorro	8	0.06531679646419819	
producer_to_prozorro	9	0.061899198654111504	
producer_to_prozorro	10	0.060185993622165784	
producer_to_prozorro	11	0.06148166507681315	
producer_to_prozorro	12	0.06268836727775298	
producer_to_prozorro	13	0.06350343977923857	
producer_to_prozorro	14	0.062210901509847344	
producer_to_prozorro	15	0.05877108946366168	
producer_to_prozorro	16	0.055021016375123494	
producer_to_prozorro	17	0.05223217889559242	
producer_to_prozorro	18	0.05074559649625219	
producer_to_prozorro	19	0.0498392096731601	
producer_to_prozorro	20	0.04919496620365539	
producer_to_prozorro	21	0.04983540657339228	
producer_to_prozorro	22	0.051914573072286835	
producer_to_prozorro	23	0.057863513984741634	
producer_to_prozorro	24	0.06963651857061677	
producer_to_prozorro	25	0.08477701379613752	
producer_to_prozorro	26	0.0976384042408879	
producer_to_prozorro	27	0.10200395079010946	
producer_to_prozorro	28	0.09874452440755055	
producer_to_prozorro	29	0.09155307713390237	
producer_to_prozorro	30	0.0829854649655558	
prozorro_to_retail	1	-0.07476478593778028	
prozorro_to_retail	2	-0.03392312633733888	
prozorro_to_retail	3	-0.27634626760712555	
prozorro_to_retail	4	0.0956104335528915	
prozorro_to_retail	5	-0.27670503024364723	
prozorro_to_retail	6	-0.22357031063621888	
prozorro_to_retail	7	-0.2955568744790299	
prozorro_to_retail	8	-0.1712031279460857	
prozorro_to_retail	9	0.0314825575192234	
prozorro_to_retail	10	-0.02732084496052369	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
hard_cheese	silpo_novus	promo_controlled	daily	I(1)	I(0)	I(0)	1793	199	60	189	49	0	Admissible for ARDL/ECM s...

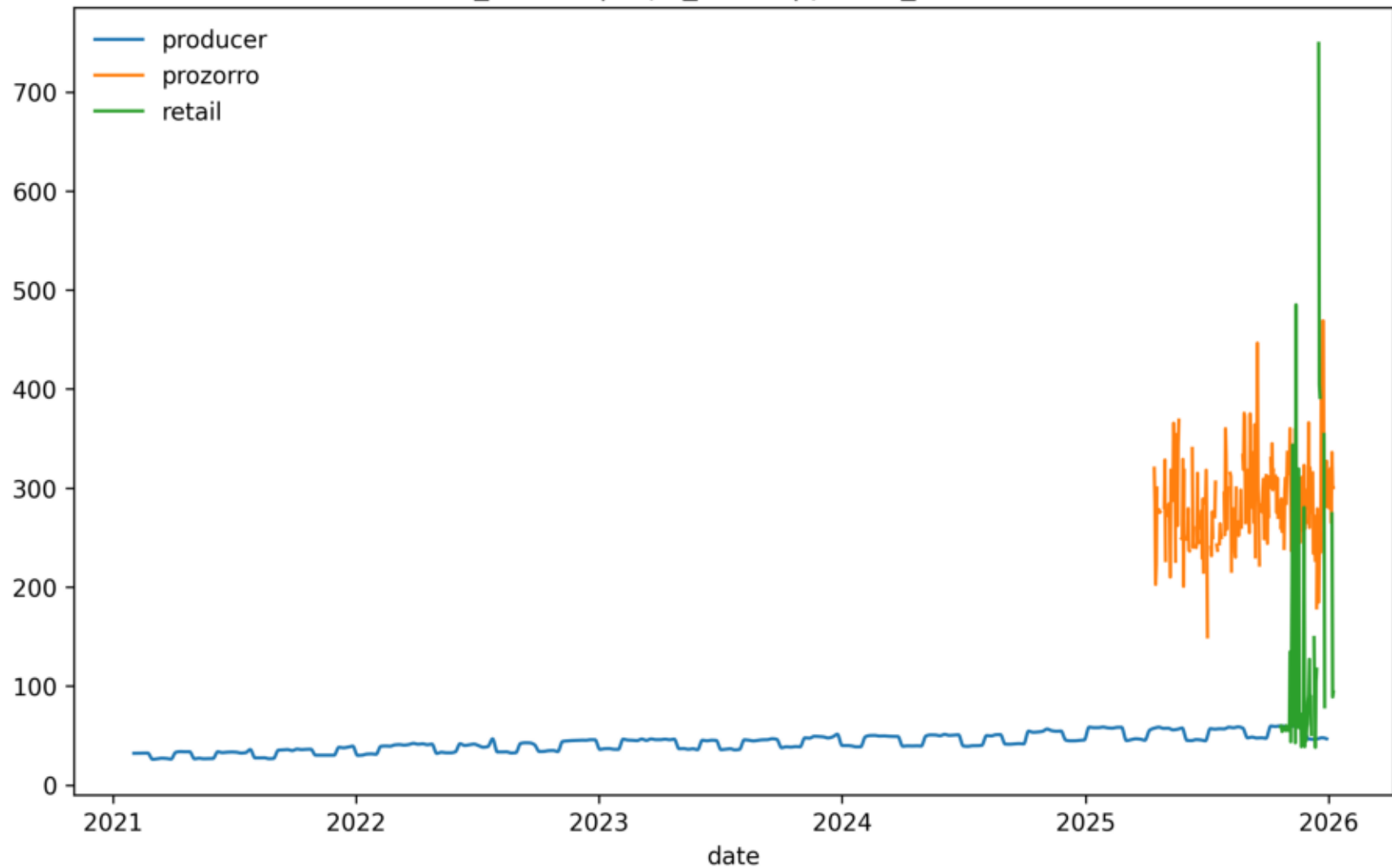
Table: NARDL_Multipliers

horizon	mult_pos	mult_neg	link
0	-1.8528869181133178	0.048797362560427326	producer_to_prozorro
1	-1.5201163099464203	0.04003356383238164	producer_to_prozorro
2	-1.5798804955409318	0.04160750480205159	producer_to_prozorro
3	-1.5691471020496899	0.04132483169956718	producer_to_prozorro
4	-1.5710747738600193	0.04137559858625983	producer_to_prozorro
5	-1.5707285722033513	0.041366481069376154	producer_to_prozorro
6	-1.5707907485481198	0.041368118536621826	producer_to_prozorro
7	-1.5707795819411023	0.04136782445447042	producer_to_prozorro
8	-1.570781587416219	0.04136787727037376	producer_to_prozorro
9	-1.5707812272414197	0.041367867784862206	producer_to_prozorro
10	-1.5707812919272812	0.04136786948841973	producer_to_prozorro
11	-1.5707812803099759	0.04136786918246804	producer_to_prozorro
12	-1.5707812823963945	0.041367869237415665	producer_to_prozorro
13	-1.5707812820216824	0.04136786922754731	producer_to_prozorro
14	-1.5707812820889793	0.04136786922931962	producer_to_prozorro
15	-1.570781282076893	0.04136786922900132	producer_to_prozorro
16	-1.5707812820790636	0.04136786922905849	producer_to_prozorro
17	-1.5707812820786737	0.04136786922904822	producer_to_prozorro
18	-1.5707812820787437	0.041367869229050065	producer_to_prozorro
19	-1.5707812820787312	0.04136786922904973	producer_to_prozorro
20	-1.5707812820787335	0.041367869229049795	producer_to_prozorro
0	1.5649515421229236	1.8518691920707206	prozorro_to_retail
1	1.0255407965328427	1.2135630754001092	prozorro_to_retail
2	1.2114660118678442	1.4335757524961257	prozorro_to_retail
3	1.1473809227590772	1.357741326319171	prozorro_to_retail
4	1.1694699042814531	1.3838800937279156	prozorro_to_retail
5	1.161856228284721	1.3748705291265542	prozorro_to_retail
6	1.164480525929431	1.3779759645526695	prozorro_to_retail
7	1.1635759775705323	1.3769055766246763	prozorro_to_retail
8	1.1638877591762644	1.3772745201572834	prozorro_to_retail
9	1.163780293645299	1.3771473519347233	prozorro_to_retail
10	1.1638173350875565	1.3771911845416125	prozorro_to_retail
11	1.1638045675663597	1.3771760762276954	prozorro_to_retail
12	1.1638089683020647	1.3771812837926936	prozorro_to_retail
13	1.1638074514473482	1.3771794888383817	prozorro_to_retail
14	1.1638079742799803	1.3771801075269594	prozorro_to_retail
15	1.1638077940689435	1.3771798942760964	prozorro_to_retail
16	1.1638078561844576	1.3771799677798464	prozorro_to_retail
17	1.1638078347743535	1.3771799424444213	prozorro_to_retail
18	1.1638078421540325	1.3771799511770881	prozorro_to_retail

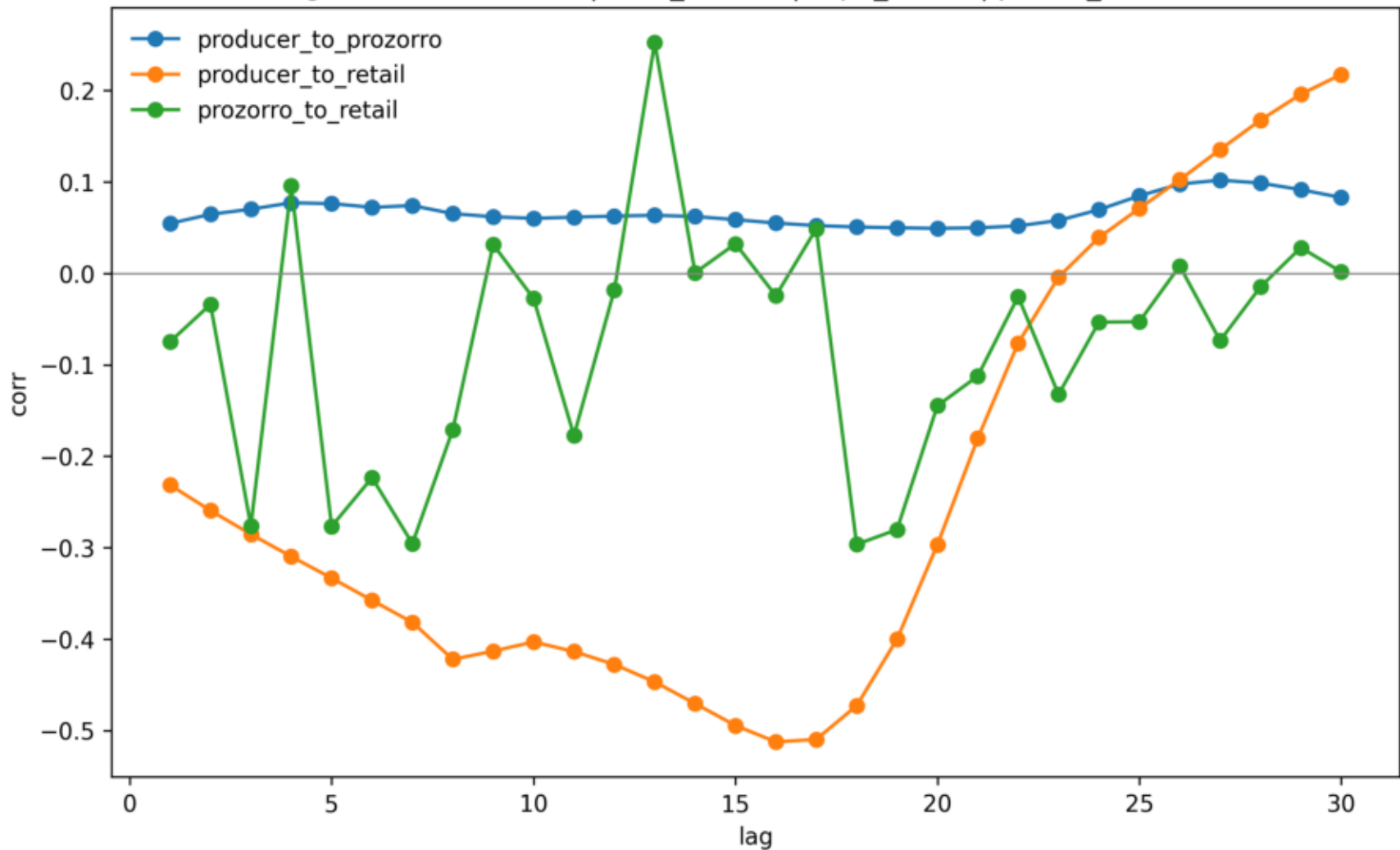
Table: VECM_IRF

No rows

hard_cheese | silpo_novus | promo_controlled



Lag-Correlation Profile | hard_cheese | silpo_novus | promo_controlled



NARDL Dynamic Multipliers | hard_cheese | silpo_novus | promo_controlled

